

FairRoute++: Quantifying the Fairness Disconnect Between Routing and Prediction

Parth Thaker¹, Hiral Chauhan², Dr. Narendrasinh Chauhan³ ¹M.Tech Student, A.D. Patel Institute of Technology, The CVM University

thakerparth12@gmail.com ²M.Tech Student, A.D. Patel Institute of Technology, The CVM University

hiralchauhan@gmail.com ³Professor and Head of Department, A.D. Patel Institute of Technology, The CVM University
head.it@adit.ac.in

Abstract—Fairness-aware learning has moved from a nice-to-have to a hard requirement wherever machine learning touches consequential decisions. A growing line of work on Mixture-of-Experts (MoE) routing takes a shortcut, though: it treats a fair router as proof of a fair model. We show that shortcut can mislead. Routing assignments can be pushed to near-independence from a sensitive attribute while the model’s final output keeps a striking amount of demographic signal intact – a gap we term the *Fairness Disconnect*. We propose FairRoute++, a sparse-routing framework that regularizes latent features, routing decisions, and predictions jointly through multi-stage adversarial training rather than debiasing the router alone. To make the gap measurable, we introduce the *Fairness Disconnect Index (FDI)*, built from routing attenuation and residual prediction leakage. Six tabular benchmarks – income prediction, recidivism, credit scoring, healthcare, and a controlled synthetic setting – all show the same behavior: driving routing mutual information toward zero barely dents prediction-level leakage. We then ask whether this is a tabular quirk. It is not. Repeating the protocol on two vision datasets (CelebA, FairFace) with a convolutional MoE, and on a text classification benchmark, reproduces the disconnect under a different modality and architecture. Seven classical baselines, a prediction-level mitigation study, two independent mutual-information estimators, and a component ablation round out the evidence. Across every angle we tried, one conclusion holds: routing fairness and prediction fairness are distinct properties, and mistaking one for the other overstates how fair a system actually is.

Index Terms—Mixture-of-Experts, Sparse routing, FairRoute++, Fairness Disconnect, Fairness Disconnect Index, Adversarial learning, Prediction leakage, Mutual information, Explainable AI.

I. INTRODUCTION

Machine learning now sits upstream of decisions that used to be made by people alone – who gets a loan, who gets flagged as high-risk, who gets a callback. Once a model shapes outcomes like these, demographic fairness stops being a nice-to-have and becomes part of the specification [6], [16]. A large body of work has responded by building systems that suppress biased correlations without giving up too much accuracy.

Three broad strategies dominate this space: pre-processing, which reshapes the training data itself; in-processing, which bakes fairness constraints into the optimization; and post-processing, which patches predictions after training [1], [2], [5]. The in-processing camp leans heavily on adversarial

learning, where a discriminator tries to guess the sensitive attribute from a representation while the encoder is pushed, via the same loss, to make that guessing hard [4], [23]. This recipe works well on ordinary architectures. Sparse routing has largely sat outside its reach.

Sparse Mixture-of-Experts (MoE) models have become popular for a simple reason: they scale capacity without paying the full compute bill, since only a handful of experts fire per input [13]–[15]. Because the router decides which experts see which inputs, it looks like the natural lever for fairness – and several recent papers pull exactly that lever, regularizing or adversarially debiasing the router directly. The implicit bet is straightforward: strip the sensitive attribute out of routing, and the rest of the model should follow.

That bet has barely been tested. A modern MoE is not just a router; it is a router sitting in front of dense latent features, per-expert transformations, and an aggregation step that recombines everything before prediction. Even once the router looks clean, these downstream stages can still carry demographic signal forward – which means routing-level fairness, taken alone, says very little about whether sensitive information ever actually leaves the system.

That gap sets up the question this paper is built around:

Does fair routing necessarily imply fair prediction?

Most evaluation protocols answer this implicitly by only ever checking routing statistics or latent-space properties. That is a partial view. A router can pass every fairness test on the books while the model’s final output still gives away the sensitive attribute to anyone who probes for it. When that happens, an evaluation anchored solely on routing behaviour ends up flattering the model far more than it deserves.

We name this failure mode the *Fairness Disconnect*: a discrepancy between how fair routing looks and how fair the prediction actually is. Rather than stopping at the router, we treat routing fairness as an intermediate checkpoint whose value has to be verified downstream, at the point where a decision is actually made.

Building on that framing, we introduce **FairRoute++**, which regularizes latent representations, routing assignments, and predictions together via multi-stage adversarial training, and the *Fairness Disconnect Index (FDI)*, a single diagnostic

number that tells you whether a routing-level fairness gain actually reached the output.

Six tabular benchmarks spanning income, criminal justice, credit, health, and a controlled synthetic setup form the core evidence, backed by mutual-information analysis, multiple-estimator robustness checks, a mitigation study, sensitivity sweeps, ablations, runtime profiling, and comparisons to both plain and fairness-aware baselines. We then push the question further: is this a tabular artifact? Two vision datasets and a text-classification benchmark say no – the same pattern shows up under a convolutional backbone and under a different data modality entirely.

This paper’s contributions, in short:

- We isolate the *Fairness Disconnect* and show, empirically, that fair routing does not entail fair prediction.
- We build **FairRoute++**, which jointly debiases latent features, routing, and prediction via multi-stage adversarial learning instead of treating the router as the whole problem.
- We define the *Fairness Disconnect Index (FDI)*, quantifying the gap between routing-level fairness and prediction-level leakage.
- We validate the phenomenon across six tabular benchmarks, two vision datasets, and a text-classification benchmark, with robustness analyses, mitigation studies, ablations, complexity profiling, and seven conventional baselines – evidence that prediction leakage survives statistically fair routing across modalities, not just in one corner of the design space.

II. RELATED WORK

Fairness research has spread across nearly every stage of the learning pipeline over the last decade. The resulting landscape can be sorted, roughly, into representation learning, adversarial debiasing, routing-level fairness, and information-leakage analysis. Each of these lines has made real progress at the stage it targets; none of them, as far as we can tell, checks whether that progress survives to the final prediction.

A. Fair Representation Learning

The goal here is a latent space that keeps what the task needs and drops what identifies protected groups. Zemel et al. started this thread with representations explicitly optimized to hide protected variables while staying useful for prediction [3]; the Information Bottleneck line of work later gave this trade-off a cleaner formal shape, trading task-relevant information against compression of nuisance variables [9]. Since then, mutual-information penalties, variational objectives, and disentanglement have all been tried as ways to push latent embeddings toward demographic invariance.

These methods reliably shrink bias measured *in the latent space*. What they generally assume – without checking – is that a clean representation is the whole story. In a deep network, later layers can reconstruct or exploit whatever demographic signal is left scattered across the features, even when no single feature looks incriminating on its own.

B. Adversarial Fairness

Adversarial fairness is probably the most-used recipe in this space. A discriminator tries to recover the sensitive attribute from a representation; the encoder, trained against the same signal through a gradient-reversal layer, learns to make that recovery hard [4], [23]. Because it makes few assumptions about the data distribution, this trick has been bolted onto a wide range of fairness settings.

Later variants mix the adversarial loss with classification losses or variational terms to trade accuracy against invariance more gracefully. Even so, evaluation tends to stop at representation-level statistics or standard group-fairness numbers. Whether the suppressed information quietly resurfaces further downstream is a question these papers rarely ask.

C. Fairness in Sparse Routing Architectures

Sparse MoE architectures activate a subset of experts per input, which is how they get more capacity without a proportional compute bill [13]–[15]. Since the router is what decides expert assignment, it is the obvious target for fairness interventions – and a handful of recent papers take exactly that route, pushing routing assignments toward independence from the sensitive attribute.

FairMoE and similar routing-regularization schemes are representative here [19]: they report cleaner routing statistics and improved conventional fairness metrics side by side. What they do not check, at least not systematically, is whether a debiased router actually translates into a debiased prediction. That assumption is doing a lot of unexamined work.

D. Information Leakage Analysis

A separate line of work uses information theory directly. Mutual information quantifies how much a representation still depends on a sensitive attribute; leakage analysis asks the same question with an external probe, checking whether that attribute can be recovered from internal states [8], [20]. This toolkit shows up across fairness, privacy, and representation-learning papers because it measures retained information rather than a proxy for it.

Its blind spot is scope: leakage studies almost always look at latent representations and stop there. Sparse routing introduces an extra intermediate variable – the routing assignment – whose relationship to downstream leakage has never been mapped systematically. Without that mapping, there is no way to tell whether fairness gained at the router survives the rest of the pipeline.

E. Research Gap

Routing-fairness methods have made real, measurable progress at the router and in latent representations. What none of them do is check whether that progress reaches the prediction. Leakage-analysis work, symmetrically, measures residual information in representations but rarely ties that measurement back to routing behaviour in sparse architectures.

This paper sits in that gap. We name the discrepancy the *Fairness Disconnect*, propose **FairRoute++** as a framework

that treats all three pipeline stages as first-class fairness targets, and introduce the *Fairness Disconnect Index (FDI)* so the gap can be measured rather than assumed away.

III. PROPOSED METHOD

A. Motivation

The standard argument for router-only debiasing runs like this: the router decides which expert sees which input, so if routing decisions are statistically independent of the sensitive attribute, the rest of the pipeline should inherit that independence. It is a reasonable-sounding argument, and it is why most existing evaluation protocols stop at routing statistics or adversarial routing losses.

But routing is one stage among several. A real MoE also has a shared feature extractor, per-expert transformations, and an aggregation step, all sitting between the router and the final prediction. Whatever demographic signal survives in the latent representation can ride through these stages untouched, even after the router itself has been scrubbed clean.

Figure 1 sketches this information flow. An encoder maps the input to a shared latent Z ; a routing network turns Z into assignments R . Prior routing-fairness work targets the dependence between R and the sensitive attribute S almost exclusively, on the (largely untested) premise that killing that dependence is enough.

It is not enough, because R is not what produces the final output on its own. Z keeps flowing – through the experts, through aggregation, through the prediction head – and whatever it still encodes about S can resurface there, independent of how clean R looks.

This splits fairness into two distinct questions. *Routing fairness* asks whether the assignments R leak demographic information. *Prediction fairness* asks the same question of the final output $\hat{y}$. Prior routing-aware methods optimize only the first and quietly assume the second follows. Our experiments say that assumption fails often enough to matter.

We define the *Fairness Disconnect* as exactly this: demographic information surviving in predictions despite routing that looks statistically fair. Under this view, routing fairness is a checkpoint, not a destination – its value has to be confirmed at the output, which is what motivates the Fairness Disconnect Index (FDI) defined below.

FairRoute++ follows from this framing directly. Rather than debiasing the router in isolation, it adds adversarial objectives at the latent stage, the routing stage, and the prediction stage, giving a single framework the ability to both improve fairness end-to-end and expose the disconnect when improvement at one stage fails to reach another.

B. FairRoute++ Architecture

FairRoute++ regularizes three pipeline stages at once – the latent representation, the routing assignment, and the prediction – rather than treating the router as a stand-in for the whole system. Figure 2 lays out the full architecture.

Given $x \in \mathbb{R}^d$, a shared encoder produces

$$z = f_\theta(x), \quad (1)$$

where f_θ is the encoder. Unlike routing-only baselines, FairRoute++ tracks the demographic content of z directly, not just what happens to it after routing.

A softmax router turns the latent into a distribution over experts,

$$r = \text{Softmax}(W_r z + b_r), \quad (2)$$

with $r = [r_1, \dots, r_K]$ the routing weights for K experts.

Instead of hard-selecting one expert, FairRoute++ mixes all of them softly, weighted by r . Each expert transforms the same shared latent independently,

$$e_k = g_k(z), \quad k = 1, \dots, K, \quad (3)$$

and the outputs combine as

$$e = \sum_{k=1}^K r_k e_k. \quad (4)$$

A lightweight Feature Fusion Module then merges the shared latent back in alongside the aggregated expert output, rather than feeding the predictor from routed features alone,

$$h = \phi([z \parallel e]), \quad (5)$$

with $[\cdot \parallel \cdot]$ concatenation and $\phi(\cdot)$ a small learnable fusion network.

The prediction follows as

$$\hat{y} = \sigma(W_p h + b_p), \quad (6)$$

where $\sigma(\cdot)$ is the output activation.

That fusion step is not decorative. It mirrors how real sparse networks behave: the prediction depends on more than the routed path, so demographic information trapped in z can still shape $\hat{y}$ even after R has been made independent of S . This is precisely why routing fairness and prediction fairness need to be checked separately rather than assumed to move together.

C. Multi-stage Adversarial Learning

Most routing-fairness work stops at scrubbing demographic information from R . That helps at the router but leaves the latent representation and the prediction itself unconstrained. FairRoute++ instead runs three adversarial objectives in parallel, one per pipeline stage.

1) *Latent Representation Adversary*: The first adversary sits on the encoder’s output and tries to recover S from it,

$$\hat{s}_Z = A_Z(z), \quad (7)$$

with $A_Z(\cdot)$ the latent-stage classifier.

A gradient-reversal layer connects encoder and adversary. The adversary chases its own loss down; the encoder, through the same gradient (flipped), is pushed to make that chase harder. Formally,

$$\mathcal{L}_{adv}^Z = \text{CE}(\hat{s}_Z, s), \quad (8)$$

with $\text{CE}(\cdot)$ binary cross-entropy.

Figure 2. FairRoute++ Framework Overview

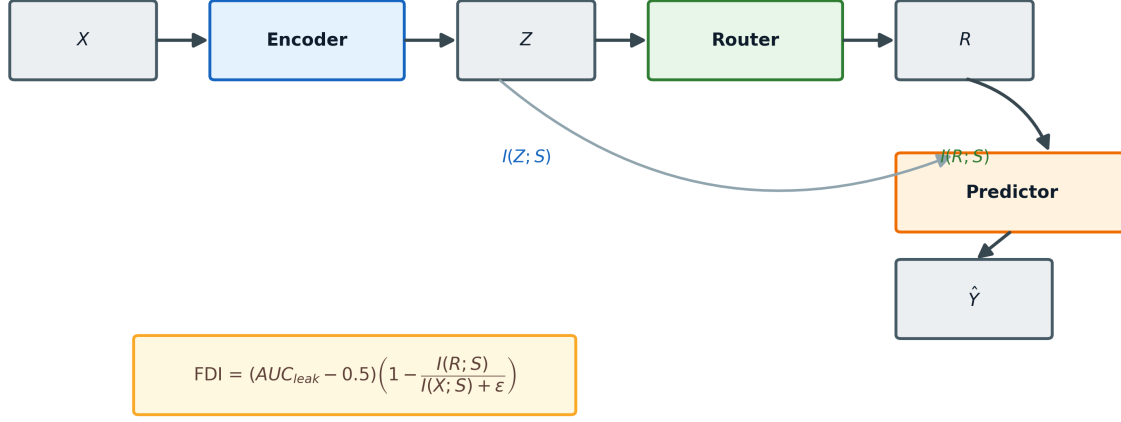


Fig. 1. Overview of the proposed FairRoute++ framework. The encoder first learns a latent representation Z , which is processed by the fairness-aware router to produce routing assignments R . While routing fairness reduces demographic information available to the router, latent representations may still propagate sensitive information to the prediction module. The proposed Fairness Disconnect Index (FDI) quantifies this discrepancy by jointly considering routing mutual information and prediction leakage.

Figure 3. FairRoute++ Architecture (with Feature Fusion Module)

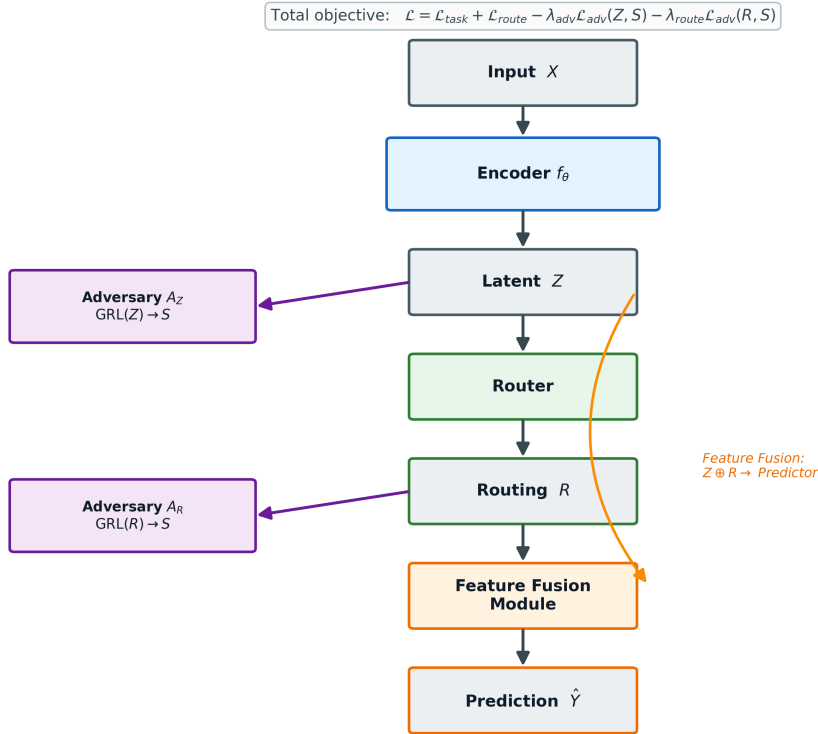


Fig. 2. Architecture of the proposed FairRoute++ framework. The encoder first learns a shared latent representation Z , which is processed by the fairness-aware router to obtain routing probabilities R . Two adversarial branches remove sensitive information from the latent representation (A_Z) and routing assignments (A_R), respectively. The routing outputs are combined with the latent representation through the proposed Feature Fusion Module before the final prediction is produced. This design enables FairRoute++ to jointly improve routing fairness while explicitly monitoring downstream prediction leakage.

2) *Routing Adversary*: The second branch operates on the routing distribution itself, trying to pull S back out of r ,

$$\hat{s}_R = A_R(r). \quad (9)$$

giving

$$\mathcal{L}_{adv}^R = \text{CE}(\hat{s}_R, s). \quad (10)$$

A gradient-reversal layer here nudges the router toward assignments that give the adversary as little to work with as possible, while still keeping routing useful for the task.

3) *Prediction Adversary*: Because demographic signal can bypass the router entirely – riding through latent features and expert outputs instead – FairRoute++ adds a third adversary directly on the final output,

$$\hat{s}_Y = A_Y(\hat{y}), \quad (11)$$

with

$$\mathcal{L}_{adv}^Y = \text{CE}(\hat{s}_Y, s). \quad (12)$$

This term stays off in the main FairRoute++ configuration and is switched on only in the mitigation study (Section V-E), where the question is whether direct pressure on the output can shrink the disconnect that the first two adversaries leave behind.

4) *Overall Training Objective*: The task loss is standard cross-entropy,

$$\mathcal{L}_{task} = \text{CE}(\hat{y}, y), \quad (13)$$

and routing regularization keeps expert usage balanced,

$$\mathcal{L}_{route}. \quad (14)$$

Putting everything together,

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{task} + \lambda_{route} \mathcal{L}_{route} \\ & - \lambda_Z \mathcal{L}_{adv}^Z - \lambda_R \mathcal{L}_{adv}^R - \lambda_Y \mathcal{L}_{adv}^Y, \end{aligned} \quad (15)$$

where $\lambda_Z, \lambda_R, \lambda_Y$ weight the three adversarial terms. $\lambda_Y = 0$ throughout the main results; it is switched on only for the mitigation variant.

This joint objective lets FairRoute++ push demographic signal down at three points simultaneously – latent, routing, prediction – without sacrificing much of the task performance the model was built for.

D. Fairness Disconnect Index

Routing fairness has quietly become the default yardstick for sparse-routing architectures, but it only tells half the story. A router can look statistically clean while the predictor downstream still encodes the sensitive attribute through latent features or expert transformations. Judging fairness from the router alone, in other words, cannot rule this out.

The *Fairness Disconnect Index (FDI)* is our attempt to close that gap with a single number that ties routing fairness to prediction leakage.

Prediction leakage comes from an external logistic-regression probe trained to guess S from $\hat{y}$. Its AUC, shifted by the chance baseline, gives a leakage score,

$$L = \text{AUC}_{leak} - 0.5, \quad (16)$$

where a larger L means more of the sensitive attribute survives in the prediction.

Routing fairness comes from $I(R; S)$, normalized against how much sensitive information the raw input carries in the first place,

$$\alpha = \frac{I(R; S)}{I(X; S) + \varepsilon}, \quad (17)$$

with ε a small constant guarding against division by zero. FDI combines the two:

$$\text{FDI} = (\text{AUC}_{leak} - 0.5) \left(1 - \frac{I(R; S)}{I(X; S) + \varepsilon} \right). \quad (18)$$

The first factor is how much demographic signal remains at the output; the second is how thoroughly routing has been scrubbed relative to the input. Multiplying them isolates exactly the failure mode we care about: leakage that persists *despite* routing looking fair.

A few properties fall out of this construction almost for free. FDI hits zero whenever leakage is zero, no matter what routing looks like – there is nothing to disconnect from if there is no leakage to begin with. If routing itself remains dependent on S , the attenuation factor shrinks, which stops a model that never achieved routing fairness from scoring an inflated disconnect. And the score only climbs when both conditions line up together: clean-looking routing, leaky prediction. That is the definition of Fairness Disconnect, encoded directly into the metric.

We tried simpler alternatives – summing the two terms, or averaging them – and rejected both. A sum cannot tell a routing-unfair model apart from a prediction-leaky one; a ratio blows up once routing MI gets close to zero; a harmonic mean over-penalizes small fluctuations near the boundaries. The multiplicative form avoids all three problems while staying bounded and easy to read.

Low FDI says routing gains are showing up where they should – in the prediction. High FDI is a warning sign: the router looks fine, but that appearance is not being backed up downstream.

IV. EXPERIMENTAL SETUP

This section lays out the datasets, metrics, implementation choices, and protocol behind the FairRoute++ evaluation. Every experiment, unless flagged otherwise, uses the same training configuration across datasets so that differences in the results trace back to the fairness mechanism itself rather than to per-dataset tuning.

A. Benchmark Datasets

Six widely used fairness benchmarks anchor the tabular study, spanning criminal justice, income, credit, health, and a synthetic setting built for controlled testing.

- **Adult** predicts whether annual income exceeds \$50,000 using demographic and employment-related features. Gender is treated as the sensitive attribute.
- **COMPAS** predicts criminal recidivism risk from judicial records. Race is used as the protected attribute.
- **German Credit** evaluates loan default risk using financial and demographic variables, with age serving as the sensitive attribute.
- **Bank Marketing** predicts whether a client subscribes to a term deposit following a marketing campaign. Age is considered the protected attribute.
- **Pima Indians Diabetes** predicts diabetes diagnosis using clinical measurements. Age is used as the sensitive attribute.
- **Synthetic** is a controlled dataset generated with adjustable correlation between sensitive attributes and target labels, enabling controlled evaluation of Fairness Disconnect under different causal strengths.

Continuous features were standardized against training-split statistics, categorical variables encoded before training, and five-fold stratified cross-validation used throughout to keep variance from a single random split out of the picture.

B. Evaluation Metrics

We look at both sides of the coin: how well the model predicts, and how fair it is at each stage.

Accuracy and F1 cover predictive quality; Demographic Parity (DP) gap and Equalized Odds (EO) gap cover group fairness.

For leakage, an external logistic-regression probe attempts to recover the sensitive attribute from the model’s prediction; the resulting AUC is our Leakage AUC.

Routing fairness comes from mutual information between R and S , combined per Section III-D into the proposed FDI.

Unless stated otherwise, every number reported below is a mean $\pm$ standard deviation across five folds.

C. Implementation Details

FairRoute++ is implemented in PyTorch and optimized with Adam, using early stopping on a validation split. Seeds are fixed throughout for reproducibility, and all training runs on a single NVIDIA GPU.

Hyperparameters are listed in Table I; hardware and software configuration follows in Table II.

No dataset-specific tuning was performed beyond what Table I lists; keeping the configuration fixed across datasets is what lets us attribute performance differences to the fairness mechanism rather than to extra tuning effort. Every run – tabular, vision, or text – shares this same seed and evaluation pipeline, so results across sections stay directly comparable. Code and configuration files will be released publicly on acceptance.

TABLE I
TRAINING HYPERPARAMETERS USED THROUGHOUT ALL FAIRROUTE++ EXPERIMENTS.

Hyperparameter	Value
Latent Dimension	64
Number of Experts	4
λ_{adv}	1.0
λ_{route}	1.0
λ_{pred}	1.0
Learning Rate (Main)	0.001
Learning Rate (Adversary)	0.001
Optimizer	Adam
Weight Decay	0.0001
Batch Size	256
Epochs	50
Cross Validation Folds	5
Initial Gumbel Temperature	1.0
Final Gumbel Temperature	0.2
Gradient Clip Norm	1.0
Random Seed	42

TABLE II
EXPERIMENTAL ENVIRONMENT USED FOR ALL EXPERIMENTS.

Configuration	Value
Programming Language	Python 3.11
Framework	PyTorch 2.x
GPU	NVIDIA Tesla T4
CUDA	CUDA Enabled
Operating System	Linux
Optimizer	Adam
Cross Validation	5-fold Stratified
Random Seed	42
Early Stopping	Enabled
Hardware	Single GPU Workstation

V. EXPERIMENTAL RESULTS

We evaluate FairRoute++ against the protocol above, starting with the core question – does routing fairness track prediction fairness? – via FDI, then moving through information flow, baseline comparisons, mitigation, robustness, ablations, and finally generalization beyond tabular data.

A. Fairness Disconnect Evaluation

Table III lays out accuracy, EO gap, and the full MI/leakage/FDI picture for all six benchmarks.

Routing mutual information falls sharply once fairness regularization kicks in, almost everywhere. Leakage AUC, though, mostly refuses to follow it down – it sits well above the chance line even where routing MI is nearly zero, which is exactly the signature the Fairness Disconnect predicts.

Adult is a clean illustration. Regularizing the router drops routing MI from 0.0171 to 0.0006 – essentially gone – while Leakage AUC barely moves, ticking up slightly from 0.6674 to 0.6711. FDI climbs from 0.0779 to 0.1676 as a result: cleaner routing, no corresponding gain at the output.

COMPAS tells much the same story: routing MI collapses toward zero, leakage stays put. German Credit pushes the effect further still. Routing MI drops substantially under regularization, yet Leakage AUC holds above 0.78 – among the

largest disconnect scores we observe anywhere in the tabular suite.

Taken together, these numbers argue that routing fairness and prediction fairness are measuring genuinely different things, not two views of the same underlying property. A protocol that checks only the router is, at best, half an audit. Zooming out across all six rows, the pattern repeats with remarkably little variation. Adversarial routing does its job on routing statistics; that job, done well, does not automatically carry over to the prediction. Several datasets show Leakage AUC sitting comfortably above chance even where routing MI is negligible – evidence that routing fairness, taken alone, is not a reliable stand-in for prediction fairness.

B. Information Flow Analysis

Section V-A shows *that* routing fairness and prediction fairness diverge; it does not show *how* demographic information travels through the model to get there. To trace that path, we track mutual information at three checkpoints – the raw input, the encoder’s latent output, and the router’s assignments.

Table IV reports these three MI values alongside routing attenuation, our summary statistic for how much demographic signal survives the trip from input to router.

Routing regularization consistently pulls routing MI down across the board. Adult, COMPAS, and Synthetic all clear 98% attenuation – the router, for practical purposes, has stopped depending on the sensitive attribute. German Credit attenuates less, simply because its raw input carries a stronger demographic signal to begin with.

None of that attenuation shows up as expected in the leakage numbers from Section V-A. Removing demographic information from routing, it turns out, does not remove it from the pipeline as a whole – it just means the information found a different route, most likely through the latent representation and expert outputs that feed the fusion stage.

This is the clearest mechanistic evidence for Fairness Disconnect we have: routing can be scrubbed almost completely while leakage persists, because routing is only one channel among several that the sensitive attribute can travel through. A fairness audit built around routing statistics alone is, structurally, blind to the other channels.

Four of six datasets – Adult, COMPAS, German, Bank – clear 96% routing attenuation, meaning demographic information is nearly gone from routing by any reasonable threshold. Leakage AUC does not care: it stays well above chance regardless, and FDI stays positive as a direct consequence. Even Synthetic and Diabetes, where attenuation is comparatively weaker, still leak measurably. The upshot across all six is unambiguous: scrubbing routing decisions is not sufficient on its own, and any evaluation protocol that treats it as sufficient is measuring the wrong stage of the pipeline. A sweep over the Synthetic dataset’s causal-strength knob adds a further check. FDI tracks that knob almost monotonically (Figure 3): weak causal dependence between S and Y yields small FDI, stronger dependence yields larger FDI. That responsiveness matters – it says the metric is picking up on something real in the data-generating process, not just noise from a particular optimization run.

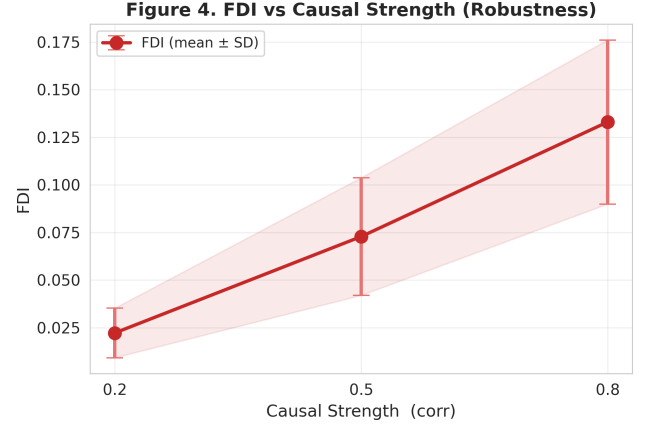


Fig. 3. Robustness of the proposed Fairness Disconnect Index (FDI) under different levels of causal dependence between the sensitive attribute and the prediction target. Error bars represent one standard deviation over repeated experiments. As the causal relationship becomes stronger, the measured FDI consistently increases, indicating that the proposed metric remains sensitive to demographic information leakage across different correlation strengths.

At the low end of the sweep, little sensitive information has any route through the pipeline to begin with, so FDI stays small almost by construction. As dependence strengthens, more of that signal survives in Z and in $\hat{y}$, and the routing/prediction gap widens visibly.

The net effect is a metric that stays interpretable across very different causal regimes – useful evidence that FDI is tracking demographic leakage specifically, rather than some artifact of the training procedure.

C. Statistical Validation of Fairness Disconnect

A pattern across datasets is suggestive; it is not proof. To rule out sampling noise as the explanation, we run a one-sample t -test on the per-fold Leakage AUC values against the null $AUC = 0.5$, and report Cohen’s d alongside it for effect size.

Table V lays out the results: the null is rejected everywhere, with p -values small enough and effect sizes large enough that chance is not a credible explanation. Prediction-level leakage holds up even where routing MI has been driven close to zero.

This closes the loop on the disconnect claim statistically: fair-looking routing is not accompanied, reliably, by fair predictions, and that gap is not a fluke of any single dataset.

Every dataset in Table V rejects H_0 at the 5% level, and the effect sizes are not marginal – they are large by any conventional reading of Cohen’s d . That combination is what lets us say the observed leakage is both statistically real and practically substantial, not a rounding artifact that happens to clear a significance threshold.

D. Baseline FDI Evaluation

Is FDI picking up something specific to FairRoute++’s architecture, or something more general about fairness-aware learning? To find out, we ran seven conventional models – Logistic Regression (LR), Random Forest (RF), a Multi-Layer Perceptron (MLP), Gradient Boosting (GBDT), XGBoost

TABLE III

OVERALL PERFORMANCE OF FAIRROUTE++ ACROSS ALL BENCHMARK DATASETS. RESULTS ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER FIVE-FOLD CROSS-VALIDATION. INPUT MI DENOTES THE MUTUAL INFORMATION BETWEEN THE INPUT FEATURES AND THE SENSITIVE ATTRIBUTE, WHILE ROUTING MI MEASURES DEMOGRAPHIC INFORMATION RETAINED IN ROUTING ASSIGNMENTS. LEAKAGE AUC IS COMPUTED USING AN EXTERNAL LOGISTIC REGRESSION PROBE TRAINED ON THE FINAL MODEL PREDICTIONS.

Dataset	Input MI	Latent MI	Routing MI	Leakage AUC	FDI	EO Gap	Accuracy
Synthetic	0.033 $\pm$ 0.018	0.000 $\pm$ 0.000	0.007 $\pm$ 0.003	0.584 $\pm$ 0.043	0.063 $\pm$ 0.046	0.057 $\pm$ 0.022	0.969 $\pm$ 0.010
COMPAS	0.046 $\pm$ 0.005	0.019 $\pm$ 0.014	0.001 $\pm$ 0.002	0.661 $\pm$ 0.023	0.156 $\pm$ 0.025	0.223 $\pm$ 0.040	0.681 $\pm$ 0.013
Adult	0.032 $\pm$ 0.003	0.032 $\pm$ 0.003	0.000 $\pm$ 0.000	0.655 $\pm$ 0.007	0.154 $\pm$ 0.009	0.032 $\pm$ 0.018	0.829 $\pm$ 0.004
German	0.181 $\pm$ 0.070	0.117 $\pm$ 0.068	0.006 $\pm$ 0.004	0.889 $\pm$ 0.066	0.371 $\pm$ 0.080	0.239 $\pm$ 0.095	0.743 $\pm$ 0.014
Bank	0.653 $\pm$ 0.001	0.500 $\pm$ 0.041	0.002 $\pm$ 0.003	0.979 $\pm$ 0.009	0.478 $\pm$ 0.008	0.030 $\pm$ 0.016	0.916 $\pm$ 0.003
Diabetes	0.526 $\pm$ 0.017	0.506 $\pm$ 0.046	0.129 $\pm$ 0.025	0.983 $\pm$ 0.008	0.364 $\pm$ 0.022	0.331 $\pm$ 0.072	0.753 $\pm$ 0.026

TABLE IV

INFORMATION FLOW ACROSS THE PROPOSED FAIRROUTE++ FRAMEWORK. ROUTING ATTENUATION MEASURES THE PERCENTAGE REDUCTION IN DEMOGRAPHIC INFORMATION FROM THE INPUT REPRESENTATION TO ROUTING ASSIGNMENTS.

Dataset	Input MI	Latent MI	Routing MI	Routing Attenuation (%)	Leakage AUC	FDI
Synthetic	0.0326	0.0000	0.0074	77.4	0.5837	0.0627
COMPAS	0.0463	0.0194	0.0014	97.0	0.6610	0.1564
Adult	0.0320	0.0317	0.0002	99.3	0.6549	0.1538
German	0.1814	0.1170	0.0063	96.5	0.8889	0.3708
Bank	0.6529	0.4996	0.0020	99.7	0.9790	0.4776
Diabetes	0.5260	0.5061	0.1293	75.4	0.9834	0.3644

TABLE V

STATISTICAL VALIDATION OF PREDICTION LEAKAGE. A ONE-SAMPLE t -TEST EVALUATES WHETHER THE OBSERVED LEAKAGE AUC IS SIGNIFICANTLY GREATER THAN RANDOM GUESSING ($AUC = 0.5$). COHEN’S d QUANTIFIES EFFECT SIZE, WHILE THE CONFIDENCE INTERVAL IS COMPUTED OVER THE FIVE CROSS-VALIDATION FOLDS.

Dataset	Leakage AUC	95% CI	p -value	Cohen’s d	Reject H_0
Synthetic	0.5837 $\pm$ 0.0428	[0.5306, 0.6368]	0.0060	1.956	Yes
COMPAS	0.6610 $\pm$ 0.0231	[0.6324, 0.6897]	<0.001	6.983	Yes
Adult	0.6549 $\pm$ 0.0072	[0.6459, 0.6638]	<0.001	21.417	Yes
German	0.8889 $\pm$ 0.0663	[0.8066, 0.9713]	<0.001	5.863	Yes
Bank	0.9790 $\pm$ 0.0086	[0.9684, 0.9897]	<0.001	55.699	Yes
Diabetes	0.9834 $\pm$ 0.0084	[0.9730, 0.9938]	<0.001	57.874	Yes

(XGB), LightGBM (LGBM), and Exponentiated Gradient with Equalized Odds (ExpGrad+EO) – through the identical cross-validation splits.

None of these models has a router, so routing MI is not defined for them; FDI collapses to a direct function of prediction leakage in this comparison, using the same probing setup as everywhere else in the paper.

Table VI reports Leakage AUC, FDI, and a one-sample t -test against chance ($AUC = 0.5$) for each model on each dataset; Wilcoxon signed-rank tests, run in parallel, agreed with every t -test call and are omitted here for space.

Leakage, plainly, has nothing special to do with FairRoute++’s architecture. Linear models, tree ensembles, and plain neural nets all show positive, statistically significant FDI under this same protocol – 30 of 35 model-dataset pairs reject the chance-guessing null outright. ExpGrad+EO is the interesting exception: it pulls Leakage AUC toward 0.5 on Bank, German, and COMPAS, and loses significance on exactly those three, which is itself a useful sanity check – the test is not rubber-stamping every model regardless of what it does. Where ExpGrad+EO does not fully neutralize leakage, though, the finding stands on its own: fairness-aware post-processing does not guarantee the sensitive attribute has

actually left the prediction.

Read together, this places Fairness Disconnect well outside the sparse-routing box. Hidden leakage turns up across model families sharing nothing architecturally, which means conventional fairness metrics – taken alone – miss something general about learned predictors, not something peculiar to MoE routing.

E. Mitigation Experiment

Everything so far shows leakage surviving fair-looking routing. The obvious next question: can a targeted fix actually shrink it?

We test this with a prediction-level adversarial mitigation module – an auxiliary adversary attached directly to $\hat{y}$, trained to minimize how much of S it can recover while the model still tries to keep its accuracy. Where the routing adversary works upstream, this one goes straight at the residual leakage that slipped past it.

Table VII sets the original FairRoute++ model against this mitigation variant on Adult and COMPAS, reporting Leakage AUC, FDI, EO gap, and accuracy for both. The results here are, frankly, a little humbling. On Adult, the prediction adversary nudges FDI and EO gap down slightly, at a small

TABLE VI
FAIRNESS DISCONNECT INDEX ACROSS CONVENTIONAL MACHINE LEARNING BASELINES, WITH A ONE-SAMPLE t -TEST OF LEAKAGE AUC AGAINST CHANCE ($AUC = 0.5$). RESULTS ARE MEAN $\pm$ STANDARD DEVIATION OVER FIVE-FOLD CROSS-VALIDATION.

Dataset	Model	Leakage AUC	FDI	t -test p	Reject H_0
COMPAS	LR	0.681 $\pm$ 0.013	0.181 $\pm$ 0.013	≤ 0.001	Yes
	RF	0.625 $\pm$ 0.012	0.125 $\pm$ 0.012	≤ 0.001	Yes
	MLP	0.678 $\pm$ 0.014	0.178 $\pm$ 0.014	≤ 0.001	Yes
	GBDT	0.672 $\pm$ 0.013	0.172 $\pm$ 0.013	≤ 0.001	Yes
	XGB	0.653 $\pm$ 0.010	0.153 $\pm$ 0.010	≤ 0.001	Yes
	LGBM	0.660 $\pm$ 0.010	0.160 $\pm$ 0.010	≤ 0.001	Yes
	ExpGrad+EO	0.547 $\pm$ 0.060	0.047 $\pm$ 0.060	0.077	No
Adult	LR	0.592 $\pm$ 0.003	0.092 $\pm$ 0.003	≤ 0.001	Yes
	RF	0.594 $\pm$ 0.006	0.094 $\pm$ 0.006	≤ 0.001	Yes
	MLP	0.610 $\pm$ 0.002	0.110 $\pm$ 0.002	≤ 0.001	Yes
	GBDT	0.613 $\pm$ 0.002	0.113 $\pm$ 0.002	≤ 0.001	Yes
	XGB	0.607 $\pm$ 0.004	0.107 $\pm$ 0.004	≤ 0.001	Yes
	LGBM	0.609 $\pm$ 0.003	0.109 $\pm$ 0.003	≤ 0.001	Yes
	ExpGrad+EO	0.526 $\pm$ 0.006	0.026 $\pm$ 0.006	≤ 0.001	Yes
German	LR	0.633 $\pm$ 0.049	0.133 $\pm$ 0.049	0.002	Yes
	RF	0.658 $\pm$ 0.050	0.158 $\pm$ 0.050	0.001	Yes
	MLP	0.621 $\pm$ 0.071	0.121 $\pm$ 0.071	0.009	Yes
	GBDT	0.650 $\pm$ 0.048	0.150 $\pm$ 0.048	0.001	Yes
	XGB	0.641 $\pm$ 0.048	0.141 $\pm$ 0.048	0.001	Yes
	LGBM	0.632 $\pm$ 0.046	0.132 $\pm$ 0.046	0.002	Yes
	ExpGrad+EO	0.493 $\pm$ 0.058	-0.007 $\pm$ 0.058	0.595	No
Bank	LR	0.507 $\pm$ 0.019	0.007 $\pm$ 0.019	0.230	No
	RF	0.507 $\pm$ 0.005	0.007 $\pm$ 0.005	0.011	Yes
	MLP	0.526 $\pm$ 0.004	0.026 $\pm$ 0.004	≤ 0.001	Yes
	GBDT	0.517 $\pm$ 0.009	0.017 $\pm$ 0.009	0.007	Yes
	XGB	0.513 $\pm$ 0.010	0.013 $\pm$ 0.010	0.020	Yes
	LGBM	0.514 $\pm$ 0.011	0.014 $\pm$ 0.011	0.026	Yes
	ExpGrad+EO	0.503 $\pm$ 0.005	0.003 $\pm$ 0.005	0.129	No
Diabetes	LR	0.727 $\pm$ 0.071	0.227 $\pm$ 0.071	0.001	Yes
	RF	0.747 $\pm$ 0.042	0.247 $\pm$ 0.042	≤ 0.001	Yes
	MLP	0.761 $\pm$ 0.053	0.261 $\pm$ 0.053	≤ 0.001	Yes
	GBDT	0.712 $\pm$ 0.050	0.212 $\pm$ 0.050	≤ 0.001	Yes
	XGB	0.687 $\pm$ 0.048	0.187 $\pm$ 0.048	0.001	Yes
	LGBM	0.693 $\pm$ 0.047	0.193 $\pm$ 0.047	≤ 0.001	Yes
	ExpGrad+EO	0.480 $\pm$ 0.055	-0.020 $\pm$ 0.055	0.765	No

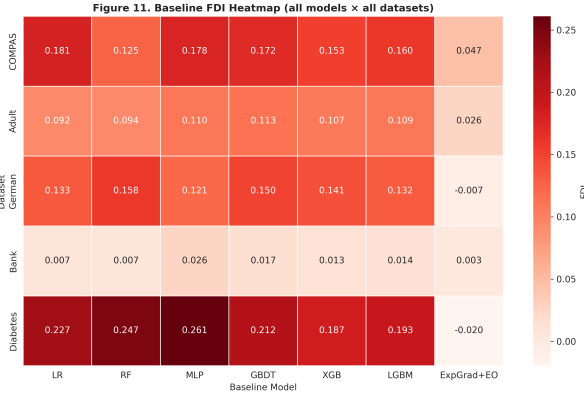


Fig. 4. Average Fairness Disconnect Index (FDI) across conventional machine learning baselines. Higher values indicate stronger residual demographic information within model predictions.

accuracy cost – a modest win. On COMPAS, Equalized Odds improves more visibly, but at a bigger accuracy cost, and FDI barely budes. One adversary bolted onto one stage is not, apparently, a reliable fix.

The likely reading: residual leakage does not live in one place, so knocking on the prediction door alone will not always close it. Fairness Disconnect looks like a distributed problem, spread across the pipeline, which means single-stage patches

TABLE VII
MITIGATION EXPERIMENT COMPARING THE ORIGINAL FAIRROUTE++ MODEL WITH AN ADDITIONAL PREDICTION-LEVEL ADVERSARIAL MITIGATION MODULE. RESULTS ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER FIVE-FOLD CROSS-VALIDATION.

Dataset	Condition	Leakage AUC	FDI	EO Gap
Adult	Baseline	0.654 $\pm$ 0.005	0.154 $\pm$ 0.005	0.042 $\pm$ 0.005
	Mitigation (+ A_Y)	0.653 $\pm$ 0.010	0.153 $\pm$ 0.010	0.024 $\pm$ 0.005
COMPAS	Baseline	0.657 $\pm$ 0.017	0.155 $\pm$ 0.018	0.231 $\pm$ 0.018
	Mitigation (+ A_Y)	0.663 $\pm$ 0.018	0.157 $\pm$ 0.025	0.182 $\pm$ 0.018

– whether at the router or at the output – are unlikely to solve it consistently.

FDI earns its keep here too, beyond just flagging the problem: it gives us a number to check whether a proposed fix actually worked, rather than trusting that it should have.

Figure 8. Mitigation Experiment — Does Adding A_Y on Y Reduce FDI?

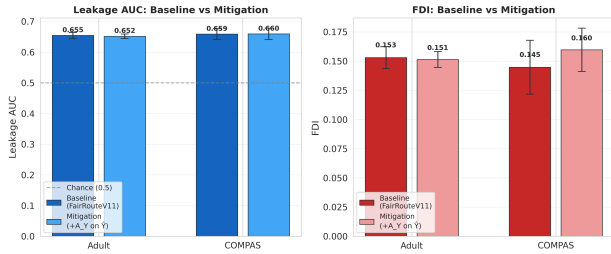


Fig. 5. Comparison of the baseline FairRoute++ model and the prediction-level mitigation strategy. While Equalized Odds improves on both datasets, reductions in the proposed Fairness Disconnect Index remain modest, indicating that prediction leakage persists despite additional adversarial regularization.

F. Mutual Information Robustness Analysis

MI estimators are notoriously estimator-dependent, especially in high dimensions – classifier-based and neural estimators in particular can run optimistic on complex feature spaces. To make sure Fairness Disconnect is not an artifact of our specific MI estimator, we cross-check the classifier-based estimator used elsewhere in this paper against a non-parametric k -nearest-neighbor (kNN) estimator.

Bank and Diabetes get the closest look here, since they show the largest leakage in the tabular suite. For each, we estimate MI independently at the input, latent, and routing stages under both estimators, alongside Leakage AUC as an estimator-agnostic anchor (it comes from an external probe, not from MI estimation at all).

Table VIII lays the comparison out.

At the input and routing stages, the two estimators agree almost exactly on both datasets. At the latent stage, Diabetes still agrees closely, but Bank’s two estimates diverge by a factor of twenty – a strong hint that the classifier-based estimator is running optimistic in that particular high-dimensional latent space.

Leakage AUC, notably, does not budge across this whole comparison, because it never depended on MI estimation to begin with. That gives us a way to separate the two possible worries: is the disconnect real, or is it an estimator artifact? Given that leakage stays high under an entirely independent measurement, the answer leans firmly toward real. MI estimates can wobble; the leakage they are meant to explain does not.

G. Component Ablation Study

Which piece of FairRoute++ is actually doing the work? To answer that on Adult, we compare four variants: the full model; the model with the routing adversary removed; the model with the latent adversary removed; and a vanilla sparse MoE with neither adversary, everything else held fixed.

We score all four on accuracy, F1, DP gap, EO gap, Leakage AUC, and FDI.

Both adversaries pull their weight, but not equally. Dropping only the routing adversary barely moves Leakage AUC and keeps accuracy where it was – routing debiasing on its own, in other words, is not carrying the disconnect fix.

Drop the latent adversary instead, and the picture shifts more: Leakage AUC climbs from about 0.656 to about 0.673, with FDI following it up. Latent features, this suggests, are a real conduit for demographic signal on their way to the output – arguably the more important of the two channels here.

Vanilla MoE, with both adversaries stripped out, leaks the most and shows the widest fairness gaps, all while matching the full model’s accuracy almost exactly. That last part matters: it rules out the possibility that FairRoute++’s leakage reduction is just a side effect of worse predictive performance.

So the two adversaries are complements, not substitutes – one handles routing, the other handles the latent pathway, and neither alone gets you the full reduction that combining them does, without meaningfully touching accuracy.

H. Correlation Analysis

If routing fairness were a good proxy for prediction fairness, the two should move together across runs. We test that directly by pooling cross-validation folds from every tabular benchmark and computing both Pearson and Spearman correlations between routing MI and Leakage AUC.

The Pearson coefficient here, $r = 0.319$ ($p = 0.120$), is weak and not significant – routing MI, on its own, explains only a small slice of the variation in downstream leakage. Spearman comes in a bit higher ($\rho = 0.430$, $p = 0.032$), pointing to a modest monotonic trend, but nowhere near strong enough to support treating routing fairness as a stand-in for prediction fairness.

We deliberately stop short of reporting a correlation between FDI and Leakage AUC as further evidence of the disconnect. FDI is defined as $(AUC_{leak} - 0.5) \cdot (1 - \alpha)$ – Leakage AUC is one of its two multiplicative factors by construction – so any correlation between the two is mathematically guaranteed rather than empirically discovered, and reporting it as independent support would be circular. The correlation that actually tests the hypothesis is the one above: routing MI against Leakage AUC, two quantities defined independently of each other. Its weakness is the evidence, not a footnote to it.

Read this way, the correlation analysis lines up with everything preceding it: routing-level fairness and prediction-level fairness are only loosely coupled at best, which is exactly why evaluating one cannot substitute for evaluating the other.

I. Training Stability

Fairness numbers that swing wildly with the random seed are not much of a foundation to build on. To check that FairRoute++’s do not, we retrain it five times on Adult with five different seeds, everything else held identical, and look at how much the results move.

Nothing here moves much. Accuracy hovers around 0.831 across all five seeds, F1 shows similarly tight variation, and FDI settles near 0.155 with a small standard deviation to match.

DP and EO gaps stay tight too, which says the adversarial optimization is converging to roughly the same solution regardless of initialization rather than landing in different basins

TABLE VIII
ROBUSTNESS ANALYSIS OF MUTUAL INFORMATION ESTIMATION USING CLASSIFIER-BASED MI AND KNN MI ESTIMATORS. LEAKAGE AUC IS ESTIMATOR-INDEPENDENT AND THEREFORE SERVES AS AN EXTERNAL VALIDATION OF PREDICTION LEAKAGE.

Dataset	Stage	Classifier MI	kNN MI	Ratio	Leakage AUC
Bank	Input (X)	0.6522	0.6524	1.00	0.9521
	Latent (Z)	0.4010	0.0199	20.11	0.9521
	Routing (R)	0.0000	0.0000	1.00	0.9521
Diabetes	Input (X)	0.5251	0.5216	1.01	0.9765
	Latent (Z)	0.4689	0.4317	1.09	0.9765
	Routing (R)	0.0412	0.0412	1.00	0.9765

TABLE IX
COMPONENT ABLATION STUDY ON THE ADULT DATASET. RESULTS ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER FIVE-FOLD CROSS-VALIDATION.

Variant	accuracy	f1	dp_gap	eo_gap	leakage_auc	fdi
Full FairRoute++	0.828 $\pm$ 0.004	0.556 $\pm$ 0.015	0.086 $\pm$ 0.008	0.044 $\pm$ 0.022	0.656 $\pm$ 0.004	0.156 $\pm$ 0.003
w/o Routing Adversary	0.828 $\pm$ 0.005	0.553 $\pm$ 0.023	0.086 $\pm$ 0.012	0.041 $\pm$ 0.022	0.656 $\pm$ 0.009	0.151 $\pm$ 0.011
w/o Latent Adversary	0.830 $\pm$ 0.005	0.559 $\pm$ 0.022	0.091 $\pm$ 0.012	0.047 $\pm$ 0.037	0.673 $\pm$ 0.006	0.172 $\pm$ 0.005
w/o Both (Vanilla MoE)	0.830 $\pm$ 0.004	0.562 $\pm$ 0.019	0.093 $\pm$ 0.011	0.047 $\pm$ 0.028	0.672 $\pm$ 0.005	0.161 $\pm$ 0.022

TABLE X
CORRELATION BETWEEN ROUTING MUTUAL INFORMATION AND LEAKAGE AUC, POOLED ACROSS ALL TABULAR BENCHMARK FOLDS.

Pair	Statistic	Value (p -value)
Routing MI vs. Leakage AUC	Pearson r	0.319 (0.120)
	Spearman ρ	0.430 (0.032)

TABLE XI
TRAINING STABILITY OF FAIRROUTE++ OVER FIVE INDEPENDENT RANDOM SEEDS ON THE ADULT DATASET. RESULTS ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION TOGETHER WITH THE CORRESPONDING 95% CONFIDENCE INTERVALS.

Dataset	Metric	Mean $\pm$ Std (5 seeds)	95% CI
Adult	accuracy	0.8309 $\pm$ 0.0035	[0.8265, 0.8352]
	f1	0.5578 $\pm$ 0.0109	[0.5443, 0.5713]
	dp_gap	0.0897 $\pm$ 0.0033	[0.0856, 0.0939]
	eo_gap	0.0497 $\pm$ 0.0147	[0.0314, 0.0679]
	leakage_auc	0.6548 $\pm$ 0.0061	[0.6473, 0.6624]
	fdi	0.1548 $\pm$ 0.0061	[0.1472, 0.1624]

each time. Leakage AUC, in particular, barely shifts across runs – so the leakage we keep reporting is not a quirk of one lucky (or unlucky) seed.

Put simply: FairRoute++ is not fragile to its random initialization, and the fairness story it tells is one you would get again on a different run.

J. Computational Complexity Analysis

A fairness method nobody can afford to run is not much of a method. So beyond accuracy and fairness, we also profile FairRoute++’s footprint against the classical baselines – parameter count, training time, inference latency, GPU memory – all under the same setup described in Section IV.

At roughly 22k trainable parameters, FairRoute++ is far smaller than most deep architectures, and it still delivers competitive accuracy alongside its fairness machinery. A full training run on Adult takes about 55 seconds – the extra cost of optimizing two adversarial objectives on top of the task loss.

That cost stays entirely on the training side. Inference runs at about 0.001 ms per sample, in the same ballpark as lightweight neural baselines and noticeably faster than the ensemble methods in the comparison, since the adversarial branches are training-only scaffolding that gets discarded before deployment.

Memory tells a similar story: roughly 20 MB for both training and inference, a modest footprint given how much richer the fairness diagnostics are compared to routing-only evaluation.

Put plainly, FairRoute++’s overhead is real but small, and the fairness insight it buys – catching a failure mode routing-only evaluation would miss entirely – looks like a reasonable trade for that cost.

K. Generalization Beyond Tabular Data

Everything up to this point runs on tabular benchmarks and a comparatively shallow MoE. A fair question follows immediately: is Fairness Disconnect a property of sparse routing in general, or an artifact of that specific testbed? We push on this directly, repeating the core protocol on two vision datasets and one text-classification benchmark, under a genuinely different backbone and architecture in each case.

1) *Vision Domain: CelebA and FairFace*: CelebA [24] and FairFace [25] replace the tabular encoder with a convolutional backbone feeding a conditional sparse-MoE image classifier at 64×64 resolution. Unless noted, expert count ($K=3$) and backbone size (“small”) match the tabular default; sweeps over both appear further below. We compare four variants –

TABLE XII
COMPUTATIONAL COMPLEXITY COMPARISON BETWEEN FAIRROUTE++ AND CONVENTIONAL MACHINE LEARNING BASELINES. TRAINING TIME CORRESPONDS TO ONE COMPLETE TRAINING RUN, WHILE INFERENCE LATENCY IS REPORTED PER SAMPLE.

Model	n_params	train_time_s	infer_time_ms_per_sample	gpu_mem_train_mb	gpu_mem_infer_mb
FairRoute++	22,332	54.767	0.0010	20.3	20.2
LR	6	0.030	0.0000	N/A (CPU model)	N/A (CPU model)
RF	N/A	4.259	0.0164	N/A (CPU model)	N/A (CPU model)
MLP	9,217	14.557	0.0012	N/A (CPU model)	N/A (CPU model)
GBDT	N/A	4.559	0.0013	N/A (CPU model)	N/A (CPU model)
XGB	N/A	0.239	0.0018	N/A (CPU model)	N/A (CPU model)
LGBM	N/A	0.382	0.0066	N/A (CPU model)	N/A (CPU model)
ExpGrad+EO	N/A	1.239	0.0004	N/A (CPU model)	N/A (CPU model)

Vanilla MoE, routing-regularized MoE (RoutingReg), a dual adversarial baseline with adversaries on the latent and routing stages (DualAdvMoE), and the full FairRoute++ – across both datasets.

RoutingReg is the case worth staring at. It regularizes the router and nothing else, and it works – routing MI falls to 0.08–0.13 on both datasets – yet Leakage AUC barely moves off the Vanilla-MoE baseline, sitting at 0.996–0.998 regardless. A router that looks nearly independent of the sensitive attribute sits in front of a prediction that remains almost perfectly informative about it. FairRoute++, which layers latent and prediction-stage regularization on top of routing regularization, is the only variant that pushes DP gap near zero *while also* trimming Leakage AUC somewhat – though even there, leakage stays at roughly 0.99, nowhere close to chance. None of this reads as a tabular quirk: the same disconnect shows up under a convolutional backbone on real face images, at accuracy costs consistent with the fairness–utility trade-off already visible in the ExpGrad+EO rows of Table VI.

Sweeping expert count ($K \in \{2, 3, 4\}$) and backbone size (small/medium/large) on FairFace adds a capacity dimension to this picture. Backbone capacity helps: routing attenuation climbs from 27% (small) to 82% (medium) to 76% (large), and Leakage AUC edges down correspondingly, from 0.990 to 0.936 to 0.915. It never approaches chance, though, at any capacity we tried. Expert count shows no clean monotonic trend across the same range. Bigger models attenuate the disconnect; none of them eliminate it.

A component ablation on FairFace – full FairRoute++, minus VIB compression, minus routing regularization, minus the adversarial terms, and a fully vanilla MoE – turns up one especially sharp instance of the phenomenon. Removing the adversarial objective restores strong routing attenuation (routing MI falls to 0.137) while *worsening* both Leakage AUC (0.997) and the group-fairness gaps relative to the full model. Low routing MI, here, coincides with the single worst prediction-level fairness of any variant tested – about as direct a counter-example to the router-suffices assumption as this study produces. (Because this variant retains the routing regularizer while dropping only the adversarial terms, its high attenuation is expected rather than anomalous; we flag the naming here for clarity.)

Finally, pooling routing MI against Leakage AUC and against the group-fairness gaps across all FairFace runs gives Pearson $r = 0.058$ ($p = 0.807$) for routing MI vs. Leakage AUC, and $r = 0.182/0.178$ ($p \approx 0.44$ – 0.45) against DP and EO gap respectively – no significant linear relationship in either case, across 20 runs spanning four model families. As in Section V-I, we do not report a correlation between routing MI and FDI itself as separate evidence, since that relationship follows algebraically from the FDI definition rather than from anything empirical. The routing-MI-vs-Leakage-AUC result is, if anything, the cleanest single number in this whole generalization study: a router pushed to near-zero demographic dependence carries essentially no predictive power over downstream leakage.

2) *Language Domain: Text Classification:* To check generalization along a third axis – modality rather than capacity – we repeat the same four-model comparison on a text-classification benchmark with a demographic sensitive attribute, using a transformer-encoder backbone in place of both the tabular and convolutional encoders used elsewhere.

This domain behaves differently from the first two in a way that is worth naming rather than glossing over. Input mutual information here sits at roughly 0.04–0.05 nats – an order of magnitude below what we see in vision or tabular data – so the sensitive attribute is only weakly encoded in the raw input to begin with. Routing attenuation, correspondingly, is already high (96–98%) even for Vanilla MoE, with no fairness intervention applied at all; that is the opposite of what tabular and vision data show, where an unregularized router routinely tracks the sensitive attribute closely.

Leakage AUC sits much closer to chance here too, in the 0.55–0.59 range rather than 0.95–0.99. A one-sample t -test against $AUC = 0.5$ still rejects the null for three of the four models – VanillaMoE ($p = 0.038$), RoutingReg ($p = 0.019$), and FairRoutePP ($p = 0.019$) – but DualAdvMoE falls just short of significance ($p = 0.080$), and we report that honestly rather than folding it into the pattern.

Read narrowly, this domain does not overturn anything: even where routing already looks close to independent of S , a small but statistically detectable amount of leakage remains for most models, and FDI scales down with the weaker underlying signal rather than reporting a false alarm. Read more broadly,

TABLE XIII
VISION-DOMAIN MODEL COMPARISON ON CELEBA AND FAIRFACE (5-FOLD CV, MEAN $\pm$ STD).

Dataset	Model	Accuracy	Routing MI	Leakage AUC	FDI	DP Gap
FairFace	VanillaMoE	0.569 $\pm$ 0.024	0.554 $\pm$ 0.014	0.996 $\pm$ 0.001	0.069 $\pm$ 0.011	0.246 $\pm$ 0.056
	DualAdvMoE	0.570 $\pm$ 0.030	0.562 $\pm$ 0.020	0.996 $\pm$ 0.001	0.063 $\pm$ 0.012	0.214 $\pm$ 0.266
	RoutingReg	0.581 $\pm$ 0.022	0.131 $\pm$ 0.128	0.996 $\pm$ 0.001	0.395 $\pm$ 0.100	0.247 $\pm$ 0.139
	FairRoutePP	0.380 $\pm$ 0.096	0.467$\pm$0.072	0.990$\pm$0.007	0.131 $\pm$ 0.054	0.002 $\pm$ 0.004
CelebA	VanillaMoE	0.656 $\pm$ 0.014	0.565 $\pm$ 0.032	0.998 $\pm$ 0.000	0.067 $\pm$ 0.024	0.332 $\pm$ 0.092
	DualAdvMoE	0.623 $\pm$ 0.034	0.314 $\pm$ 0.251	0.998 $\pm$ 0.002	0.258 $\pm$ 0.191	0.256 $\pm$ 0.301
	RoutingReg	0.678 $\pm$ 0.007	0.081 $\pm$ 0.080	0.998 $\pm$ 0.001	0.436 $\pm$ 0.061	0.337 $\pm$ 0.084
	FairRoutePP	0.440 $\pm$ 0.107	0.482$\pm$0.138	0.990$\pm$0.014	0.127 $\pm$ 0.095	0.000 $\pm$ 0.000

TABLE XIV
LANGUAGE-DOMAIN MODEL COMPARISON (5-FOLD CV, MEAN $\pm$ STD, MUTUAL INFORMATION IN NATS).

Model	Routing MI	Leakage AUC	FDI
VanillaMoE	0.0015 $\pm$ 0.0010	0.564 $\pm$ 0.042	0.062 $\pm$ 0.041
RoutingReg	0.0015 $\pm$ 0.0015	0.577 $\pm$ 0.041	0.074 $\pm$ 0.040
DualAdvMoE	0.0010 $\pm$ 0.0008	0.556 $\pm$ 0.048	0.055 $\pm$ 0.048
FairRoutePP	0.0007 $\pm$ 0.0005	0.594 $\pm$ 0.049	0.093 $\pm$ 0.049

it is a useful check on the metric itself – FDI tracks the actual strength of demographic signal available in a domain instead of returning a flat, domain-blind number, which is what we would want from a diagnostic rather than a fixed penalty.

Across all three domains together – tabular, vision, text – the qualitative claim holds up: routing-level independence from a sensitive attribute is not a reliable predictor of prediction-level fairness, whether the underlying leakage is large (vision, most tabular sets) or comparatively small (language, in this instance).

VI. DISCUSSION

Strip away the tables, and the finding is not complicated: routing fairness and prediction fairness are different properties, however often the literature treats them as one. Routing mutual information drops sharply under regularization in every dataset and every modality we tried; demographic information in the final prediction, more often than not, does not follow it down. FDI exists to catch precisely that gap, rather than settling for either half of the picture on its own.

The mechanism is not mysterious once you look for it. Routing is a single stop on a longer route, not the destination. An adversary can scrub the router thoroughly while upstream latent features – never directly targeted – keep carrying signal that resurfaces at the prediction layer. A router with near-zero mutual information and a leaky output are not, it turns out, mutually exclusive.

The information-flow numbers back this reading up directly. Attenuation clears 95% on several tabular datasets and even higher on some vision configurations – the router really has been cleaned. Leakage does not care. It persists regardless, which is hard to explain unless more than one channel carries demographic information through the network; routing is evidently not the only door.

Nor is this specific to FairRoute++’s architecture. Plain logistic regression, tree ensembles, ordinary neural nets – all of them leak in a statistically detectable way even when they clear standard fairness checks. That cross-model consistency is a useful sanity check in its own right: FDI is picking up something general about how fairness-aware models behave, not an idiosyncrasy of our particular router design.

Generalizing across modalities strengthens this reading rather than complicating it. A convolutional MoE on face images shows the same qualitative gap as the tabular models, at a larger scale of leakage; a transformer-based text classifier shows a smaller, weaker version of the same gap, exactly proportional to how much demographic signal the raw text carries in the first place. That domain-sensitivity is arguably the strongest evidence FDI is measuring something real – a metric that inflated equally regardless of underlying signal strength would be far less trustworthy than one that scales with it.

The mitigation experiment is a slightly humbling coda. A prediction-level adversary helps Equalized Odds on some datasets but leaves FDI mostly where it was. One intervention, at one stage, is not enough on its own – closing the disconnect for good likely means constraining latent space, router, and prediction head together, not picking a favorite and hoping.

For anyone building fairness-aware routing methods going forward, the practical upshot is straightforward: reporting routing statistics alone is an incomplete audit. Predictions can keep leaking well after the router looks clean, in every modality we tested. A protocol worth trusting reports both ends of the pipeline – what the router is doing, and what the prediction actually reveals once it reaches a decision.

FairRoute++, in that sense, is less a finished fix than an instrument. FDI does not replace demographic parity or equalized odds; it sits next to them, catching a specific and previously under-examined failure mode – routing-level progress that quietly never reaches the prediction.

VII. LIMITATIONS

FairRoute++ makes a reasonably strong case for the disconnect between routing and prediction fairness, but the study has real edges worth naming plainly.

The tabular core still leans on structured benchmarks and a comparatively shallow router; the vision and language extensions in Section V-K address the scale question directly, but neither touches the multi-billion-parameter MoE models

actually deployed in production language or vision systems, where the stakes – and possibly the mechanics – could differ further.

Leakage detection throughout relies on a single external probe – logistic regression. It is the standard choice and a defensible one, but a nonlinear probe might surface demographic signal that a linear model simply cannot see, especially in the higher-dimensional vision and language settings.

MI estimation carries its own asterisk. FDI leans on $I(R; S)$, and MI estimation in high dimensions is known to be temperamental; our classifier-based and kNN estimators mostly agree, which is reassuring, and the vision-domain kNN estimator shows some instability at the latent stage specifically – consistent with known limitations of nearest-neighbor MI estimation in high-dimensional convolutional feature spaces. “Mostly” agreeing is not the same as agreeing everywhere.

And finally, FDI diagnoses; it does not cure. The mitigation experiment in Section V makes that explicit – knocking down routing leakage does not automatically knock down prediction leakage, and building something that reliably closes the gap remains open, not something this paper claims to have solved.

VIII. FUTURE WORK

The Fairness Disconnect identified here opens up more directions than this paper closes.

Scaling further is the most obvious one: the vision and language results in Section V-K establish that the disconnect survives a change of modality and architecture, but production-scale MoE systems – the trillion-parameter language and vision models this line of routing-fairness work is ultimately aimed at – remain untested, and the disconnect could plausibly grow or shrink at that scale.

Stronger mitigation is a second, more pressing direction. The mitigation study here shows that suppressing routing leakage alone does not suppress prediction leakage; joint optimization across all three pipeline stages at once, rather than patching one after the fact, looks like the more promising path.

FDI itself need not stay confined to demographic fairness. The same construction – an attenuation factor multiplied against a residual-leakage term – could plausibly extend to privacy leakage, domain-adaptation robustness, or causal representation learning, anywhere hidden information might survive despite apparent improvement at an intermediate stage.

A more formal treatment would help too: a proper information-theoretic account of why routing-stage attenuation and prediction-stage leakage decouple, going beyond the empirical demonstration in this paper, would put Fairness Disconnect on firmer theoretical ground.

Finally, folding FDI into existing fairness-evaluation toolkits would make measuring prediction leakage alongside conventional fairness metrics a matter of habit rather than a special addition – which is, ultimately, the point.

IX. CONCLUSION

The question we started with was simple: does fair routing actually produce fair predictions? Most prior work assumes

yes and moves on. Ours says: not always, and not rarely enough to ignore.

The Fairness Disconnect Index gave us a way to put that question on firmer footing – fold prediction leakage and routing mutual information into one number and watch what happens. Across six tabular benchmarks, two vision datasets, and a text-classification benchmark, the pattern held with remarkably little variation: routing-level demographic information came down reliably under FairRoute++, but a meaningful share of it kept turning up in the final prediction regardless. Fairness measured mid-pipeline, in short, does not automatically survive to the decision layer – and that holds whether the pipeline is a shallow tabular MoE, a convolutional vision model, or a transformer text classifier.

We did not lean on one experiment to make this case. Information-flow analysis, significance testing, seven conventional baselines, a mitigation study, two independent MI estimators, a regularizer sweep, and three separate data modalities all point the same direction. The disconnect was not a fluke of one estimator, one architecture, or one training run – it kept showing up.

What we hope outlasts FairRoute++ itself is a habit: checking fairness at the router and calling the job done is not enough. Intermediate fairness and final-output fairness need separate verification, because – as this paper shows, across three different kinds of data – they can quietly drift apart.

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